Exp 9: Implementation of Subnetting in CISCO Packet Tracer Simulator

AIM :

To implement subnetting in the Cisco Packet Tracer simulator by designing multiple subnets with a suitable number of hosts, assigning static IP addresses across all the subnets, connecting the subnets through routers, and observing packet transmission across the subnets.

ALGORITHM :

1. Start the experiment.
2. Open Cisco Packet Tracer and create a new blank network topology.
3. Select the subnet mask: to give each subnet at least 5 end devices plus the switch and router, borrow bits from 192.168.1.0/24 and use a /27 mask (255.255.255.224), which gives 8 subnets with 30 usable hosts each.
4. Add the devices: two routers (2901), four switches (2960) and five PCs per switch, one LAN per switch.
5. Connect the devices with copper straight-through cables (PC ↔ switch, switch ↔ router) and connect the two routers to each other.
6. Configure each router interface from the CLI: enter `enable` → `configure terminal` → `interface <name>` → `ip address <IP> <mask>` → `no shutdown`.
7. Configure the switch ports in access mode using `switchport mode access`.
8. Assign a static IP address, subnet mask and default gateway to every PC (Desktop → IP Configuration).
9. Test connectivity: send ICMP ping / simple PDUs between PCs of the same subnet and of different subnets (subnets connected via a router), and observe the result. When the subnets are not connected through a router, hosts of different subnets cannot reach each other.
10. Observe the results in the Simulation panel (Event List / PDU List Window).
11. Stop the experiment.

PROCEDURE :

Classless IP subnetting allows subnet masks that are not just the default masks of an IP class, so a limited address space can be divided into smaller subnets and used more efficiently.

Creating the topology: In Packet Tracer, create a new network, then add routers, switches and PCs by dragging them from the device panel (bottom-left) onto the workspace, and join them by dragging cables between their ports.

Subnetting: To subnet the network address 192.168.1.0/24 so that each subnet has room for at least 5 end devices, the switch and the router, a /27 mask is used. This gives 8 subnets with 30 host addresses each.

Subnet mask	Prefix	Number of subnets	Usable hosts per subnet	Block size
255.255.255.224	/27	8	30	32

IP addressing plan:

Device	Interface / Network	IP address	PCs (PC1 – PC5)
Router R1	Gig0/0	192.168.1.1	—
Router R1	Gig0/1	192.168.2.1	—
Switch S1	Fa0/1 – LAN 1 (192.168.1.0)	—	192.168.1.11 – 192.168.1.15
Switch S1	Fa0/2 – LAN 2 (192.168.2.0)	—	192.168.2.11 – 192.168.2.15
Router R2	Fa0/0	192.168.3.1	—
Router R2	Fa0/1	192.168.4.1	—
Switch S2	Fa0/1 – LAN 3 (192.168.3.0)	—	192.168.3.11 – 192.168.3.15
Switch S2	Fa0/2 – LAN 4 (192.168.4.0)	—	192.168.4.11 – 192.168.4.15

Configuring the router (right-click the router → CLI):

```
Router> enable
Router# configure terminal
Router(config)# interface FastEthernet0/0
Router(config-if)# ip address {IP address} {subnet mask}
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface FastEthernet0/1
Router(config-if)# ip address {IP address} {subnet mask}
Router(config-if)# no shutdown
Router(config-if)# exit
```

The GigabitEthernet interfaces are configured in the same way (`interface GigabitEthernet0/0`, `ip address ...`, `no shutdown`).

Configuring the switch (right-click the switch → CLI):

```
Switch> enable
Switch# configure terminal
Switch(config)# interface FastEthernet0/1
Switch(config-if)# switchport mode access
Switch(config-if)# exit
Switch(config)# interface FastEthernet0/2
Switch(config-if)# switchport mode access
Switch(config-if)# exit
```

Configuring the PCs: Click each PC → Desktop → IP Configuration, select Static, and enter the IP address, subnet mask, default gateway (the router interface of that subnet) and DNS server. The IP address must lie in the same subnet as the router interface it is connected to.

Testing the network: Open the Command Prompt of a PC and ping another PC and the router interfaces. A successful reply shows that the network is working. Packet transmission is also observed by sending a simple PDU between hosts of different subnets in Simulation mode.

OUTPUT :

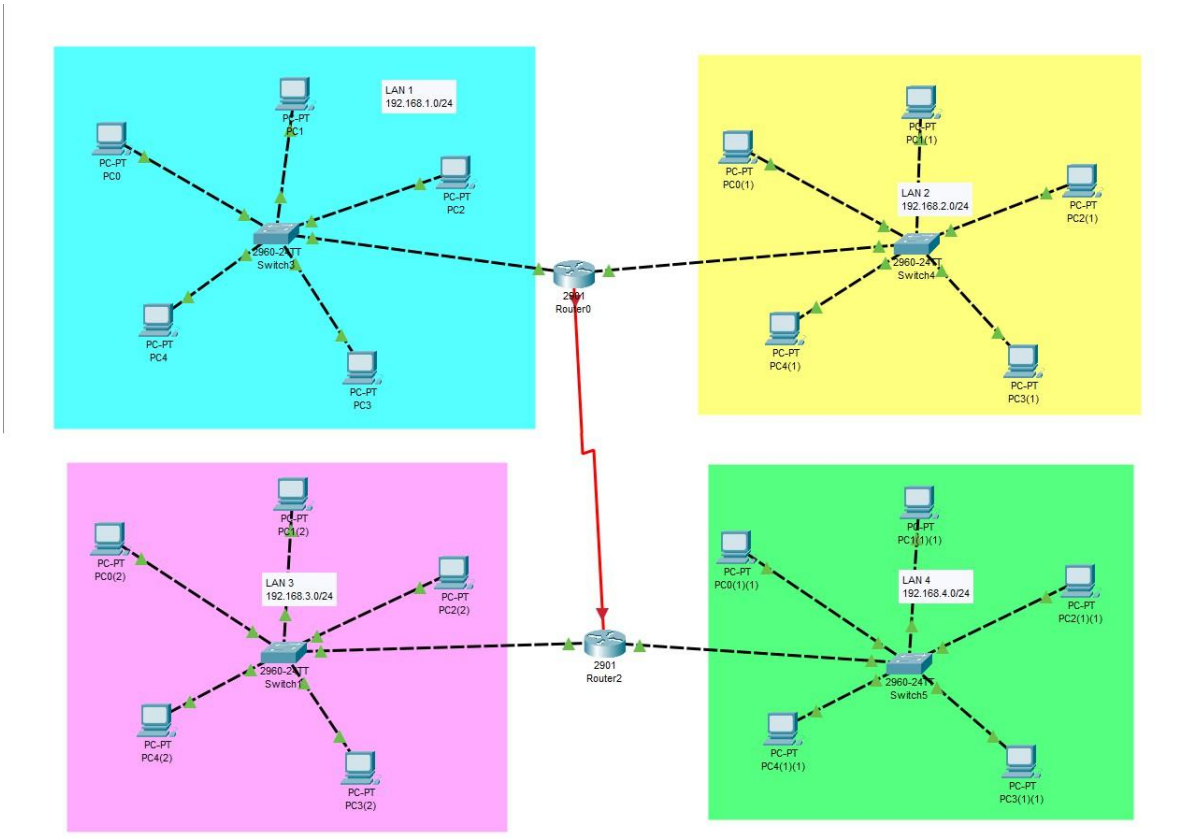


Fig 1. Network topology – four subnets (LAN 1 to LAN 4) connected through Router0 and Router2

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC4(2)	Router2	ICMP		0.000	N	12
	Successful	PC4(2)	PC2(1)(1)	ICMP		0.000	N	13
	Successful	PC0	Router0	ICMP		0.000	N	14

Fig 2. PDU list – ICMP packets transmitted successfully across the subnets (via the routers)

RESULT :

Thus, subnetting was successfully implemented in the Cisco Packet Tracer simulator. Multiple subnets were created with static IP addresses, connected through routers, and packet transmission between the subnets was verified using ICMP ping / PDUs.